AERIS Expert Review Packet

Anomaly

Source / metric	openaq / pm25
Observed / expected	52 / 10.2687
Time	2026-06-15 16:00 UTC
Location	29.8145, -95.3877
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	humidity	percent	9 / 615	57.73 to 100; mean 88.2573	100 at 2026-06-15 15:55Z; dt 5 min
asos	temperature	degC	9 / 615	22.7778 to 33.3333; mean 26.9961	26 at 2026-06-15 15:55Z; dt 5 min
asos	wind_direction	degree	9 / 615	0 to 360; mean 123.171	0 at 2026-06-15 15:55Z; dt 5 min
asos	wind_speed	m/s	9 / 615	0 to 10.2889; mean 2.5321	0 at 2026-06-15 15:55Z; dt 5 min
noaa_gfs	gh_500	m	11 / 132	5851.89 to 5926.85; mean 5897.16	5895.11 at 2026-06-15 18:00Z; dt 120 min
noaa_gfs	pbl_height	m	11 / 132	32.0322 to 1802.53; mean 391.755	187.266 at 2026-06-15 18:00Z; dt 120 min
noaa_gfs	precipitable_water	mm	11 / 132	46.5416 to 69.5435; mean 59.6164	64.0555 at 2026-06-15 18:00Z; dt 120 min
noaa_gfs	surface_pressure	hPa	11 / 132	1001.28 to 1013.9; mean 1009.35	1010.76 at 2026-06-15 18:00Z; dt 120 min
noaa_gfs	t_850	degC	11 / 132	16.785 to 19.812; mean 18.4438	17.245 at 2026-06-15 18:00Z; dt 120 min
noaa_gfs	u_10m	m/s	11 / 132	-3.00011 to 1.69078; mean -0.7036	-2.13011 at 2026-06-15 18:00Z; dt 120 min
noaa_gfs	v_10m	m/s	11 / 132	-1.42555 to 5.27924; mean 2.3038	2.07875 at 2026-06-15 18:00Z; dt 120 min
openaq	ozone	ppm	18 / 1189	-0.001 to 0.032; mean 0.0132	0.011 at 2026-06-15 16:00Z; dt 0 min
openaq	pm10	ug/m3	3 / 211	5 to 110; mean 28.0047	31 at 2026-06-15 16:00Z; dt 0 min
openaq	pm25	ug/m3	90 / 3143	-2.6 to 52; mean 7.373	52 at 2026-06-15 16:00Z; dt 0 min
openweather	cloud_cover	percent	5 / 351	1 to 100; mean 84.4615	100 at 2026-06-15 16:01Z; dt 1.9 min
openweather	humidity	percent	5 / 351	68 to 100; mean 89.8519	95 at 2026-06-15 16:01Z; dt 1.9 min
openweather	precipitation	mm/h	5 / 351	0 to 24.39; mean 1.097	0.16 at 2026-06-15 16:01Z; dt 1.9 min
openweather	pressure	hPa	5 / 351	1007 to 1016; mean 1012.14	1014 at 2026-06-15 16:01Z; dt 1.9 min
openweather	temperature	degC	5 / 351	24.22 to 33.3; mean 27.0644	25.33 at 2026-06-15 16:01Z; dt 1.9 min
openweather	wind_direction	degree	5 / 351	0 to 358; mean 163.57	285 at 2026-06-15 16:01Z; dt 1.9 min
openweather	wind_speed	m/s	5 / 351	0.07 to 5.67; mean 2.0184	1.34 at 2026-06-15 16:01Z; dt 1.9 min
purpleair	pm25	ug/m3	69 / 4985	0 to 46.2; mean 8.588	5.6 at 2026-06-15 16:00Z; dt 0 min

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
sentinel5p	s5p_aer_ai_granule_available	count	7 / 7	1 to 1; mean 1	1 at 2026-06-15 19:04Z; dt 184.6 min
sentinel5p	s5p_ch4_granule_available	count	6 / 6	1 to 1; mean 1	1 at 2026-06-15 19:04Z; dt 184.6 min
sentinel5p	s5p_co_column	mol/m^2	1 / 1	0.0259566 to 0.0259566; mean 0.026	0.0259566 at 2026-06-14 19:24Z; dt 1235.4 min
sentinel5p	s5p_co_granule_available	count	7 / 7	1 to 1; mean 1	1 at 2026-06-15 19:04Z; dt 184.6 min
sentinel5p	s5p_hcho_column	mol/m^2	1 / 1	0.000111617 to 0.000111617; mean 0.0001	0.000111617 at 2026-06-14 19:24Z; dt 1235.4 min
sentinel5p	s5p_hcho_granule_available	count	5 / 5	1 to 1; mean 1	1 at 2026-06-15 19:04Z; dt 184.6 min
sentinel5p	s5p_no2_column	mol/m^2	1 / 1	2.03572e-05 to 2.03572e-05; mean 0	2.03572e-05 at 2026-06-14 19:24Z; dt 1235.4 min
sentinel5p	s5p_no2_granule_available	count	5 / 5	1 to 1; mean 1	1 at 2026-06-15 19:04Z; dt 184.6 min
sentinel5p	s5p_o3_granule_available	count	5 / 5	1 to 1; mean 1	1 at 2026-06-15 19:04Z; dt 184.6 min
sentinel5p	s5p_so2_column	mol/m^2	1 / 1	-6.10873e-06 to -6.10873e-06; mean -0	-6.10873e-06 at 2026-06-14 19:24Z; dt 1235.4 min
sentinel5p	s5p_so2_granule_available	count	5 / 5	1 to 1; mean 1	1 at 2026-06-15 19:04Z; dt 184.6 min
tceq	co	ppm	3 / 213	0.1 to 1.1; mean 0.223	0.4 at 2026-06-15 16:00Z; dt 0 min
tceq	no2	ppb	13 / 831	-2.5 to 27; mean 5.268	7.2 at 2026-06-15 16:00Z; dt 0 min
tceq	so2	ppb	3 / 218	-0.6 to 0.6; mean -0.1307	-0.5 at 2026-06-15 16:00Z; dt 0 min

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	humidity	06-14 00Z 84.13, 06Z 88.03, 12Z 80.89, 18Z 69.98, 06-15 00Z 88.16, 06Z 92.9, 12Z 93.04, 18Z 90.65, 06-16 00Z 93.26, 06Z 97.13, 12Z 93.15, 18Z 86.17, 06-17 00Z 87.65
asos	temperature	06-14 00Z 27.49, 06Z 27.09, 12Z 28.76, 18Z 30.74, 06-15 00Z 27.65, 06Z 26.9, 12Z 25.98, 18Z 26.07, 06-16 00Z 25.96, 06Z 25.46, 12Z 25.49, 18Z 26.98, 06-17 00Z 26.48
asos	wind_direction	06-14 00Z 151.8, 06Z 124.5, 12Z 147, 18Z 176.3, 06-15 00Z 115, 06Z 124, 12Z 173.5, 18Z 106.5, 06-16 00Z 68.46, 06Z 67.17, 12Z 155.8, 18Z 106.6, 06-17 00Z 100.3
asos	wind_speed	06-14 00Z 4.055, 06Z 2.795, 12Z 3.611, 18Z 4.066, 06-15 00Z 2.229, 06Z 2.379, 12Z 2.744, 18Z 1.271, 06-16 00Z 1.533, 06Z 1.126, 12Z 3.255, 18Z 2.048, 06-17 00Z 2.925
noaa_gfs	gh_500	06-14 06Z 5923, 12Z 5911, 18Z 5926, 06-15 00Z 5921, 06Z 5922, 12Z 5902, 18Z 5893, 06-16 00Z 5886, 06Z 5890, 12Z 5872, 18Z 5865, 06-17 00Z 5855
noaa_gfs	pbl_height	06-14 06Z 759, 12Z 229.8, 18Z 1456, 06-15 00Z 448.6, 06Z 216.1, 12Z 237.8, 18Z 291.3, 06-16 00Z 147.6, 06Z 64.67, 12Z 137, 18Z 422.2, 06-17 00Z 290.8
noaa_gfs	precipitable_water	06-14 06Z 47.78, 12Z 57.95, 18Z 58.89, 06-15 00Z 60.59, 06Z 60.11, 12Z 58.53, 18Z 65.25, 06-16 00Z 61.22, 06Z 62.48, 12Z 64.15, 18Z 63.22, 06-17 00Z 55.21
noaa_gfs	surface_pressure	06-14 06Z 1010, 12Z 1011, 18Z 1012, 06-15 00Z 1010, 06Z 1011, 12Z 1011, 18Z 1011, 06-16 00Z 1008, 06Z 1009, 12Z 1008, 18Z 1008, 06-17 00Z 1005
noaa_gfs	t_850	06-14 06Z 19.65, 12Z 18.52, 18Z 17.94, 06-15 00Z 18.91, 06Z 19.24, 12Z 18.53, 18Z 17.23, 06-16 00Z 18.79, 06Z 18.4, 12Z 17.77, 18Z 17.52, 06-17 00Z 18.82
noaa_gfs	u_10m	06-14 06Z -1.733, 12Z -1.298, 18Z -0.8197, 06-15 00Z -0.9191, 06Z -0.482, 12Z -0.3766, 18Z -0.8674, 06-16 00Z -0.4467, 06Z -0.7621, 12Z 0.4799, 18Z 0.3746, 06-17 00Z -1.594

Source	Metric	Values
noaa_gfs	v_10m	06-14 06Z 4.171, 12Z 1.77, 18Z 4.332, 06-15 00Z 2.892, 06Z 2.479, 12Z 1.911, 18Z 2.624, 06-16 00Z -0.1746, 06Z 1.175, 12Z 1.674, 18Z 4.18, 06-17 00Z 0.6121
openaq	ozone	06-14 00Z 0.01563, 06Z 0.01419, 12Z 0.01556, 18Z 0.02219, 06-15 00Z 0.01393, 06Z 0.01044, 12Z 0.008887, 18Z 0.01534, 06-16 00Z 0.00795, 06Z 0.002778, 12Z 0.01004, 18Z 0.01732, 06-17 00Z 0.01571
openaq	pm10	06-14 00Z 42.33, 06Z 44.78, 12Z 33, 18Z 22.17, 06-15 00Z 33.06, 06Z 49.44, 12Z 41.22, 18Z 12.22, 06-16 00Z 20.89, 06Z 23.69, 12Z 18.33, 18Z 15, 06-17 00Z 18.67
openaq	pm25	06-14 00Z 8.891, 06Z 9.805, 12Z 7.408, 18Z 8.402, 06-15 00Z 9.636, 06Z 9.642, 12Z 7.469, 18Z 4.56, 06-16 00Z 7.97, 06Z 7.125, 12Z 4.252, 18Z 2.61, 06-17 00Z 2.805
openweather	cloud_cover	06-14 00Z 2.3, 06Z 9, 12Z 45.19, 18Z 81.5, 06-15 00Z 95.83, 06Z 99.86, 12Z 99.13, 18Z 99.93, 06-16 00Z 99.57, 06Z 99.33, 12Z 99.9, 18Z 100, 06-17 00Z 98.67
openweather	humidity	06-14 00Z 82.8, 06Z 88.08, 12Z 84.92, 18Z 73.03, 06-15 00Z 89, 06Z 92.72, 12Z 93.48, 18Z 94.13, 06-16 00Z 92.93, 06Z 95.23, 12Z 94.17, 18Z 91.79, 06-17 00Z 90.29
openweather	precipitation	06-14 00Z 0, 06Z 0, 12Z 0.1485, 18Z 0.1917, 06-15 00Z 0.203, 06Z 0.19, 12Z 3.307, 18Z 4.336, 06-16 00Z 0.7973, 06Z 1.585, 12Z 1.259, 18Z 0.7517, 06-17 00Z 0.008095
openweather	pressure	06-14 00Z 1013, 06Z 1013, 12Z 1015, 18Z 1013, 06-15 00Z 1013, 06Z 1013, 12Z 1014, 18Z 1012, 06-16 00Z 1012, 06Z 1011, 12Z 1011, 18Z 1009, 06-17 00Z 1008
openweather	temperature	06-14 00Z 27.86, 06Z 27.41, 12Z 28.83, 18Z 31.85, 06-15 00Z 27.99, 06Z 27.09, 12Z 26.29, 18Z 25.43, 06-16 00Z 25.95, 06Z 25.36, 12Z 25.52, 18Z 26.41, 06-17 00Z 26.5
openweather	wind_direction	06-14 00Z 161.1, 06Z 140.3, 12Z 170, 18Z 182, 06-15 00Z 151.6, 06Z 164.7, 12Z 187.4, 18Z 163.7, 06-16 00Z 125.5, 06Z 198.5, 12Z 184.4, 18Z 165, 06-17 00Z 111
openweather	wind_speed	06-14 00Z 3.136, 06Z 1.735, 12Z 2.405, 18Z 3.247, 06-15 00Z 2.106, 06Z 2.097, 12Z 1.986, 18Z 1.815, 06-16 00Z 1.037, 06Z 1.571, 12Z 1.688, 18Z 2.229, 06-17 00Z 1.919
purpleair	pm25	06-14 00Z 10.04, 06Z 11.39, 12Z 8.916, 18Z 10.96, 06-15 00Z 11.85, 06Z 10.83, 12Z 7.784, 18Z 6.434, 06-16 00Z 9.419, 06Z 9.409, 12Z 4.718, 18Z 5.139, 06-17 00Z 5.02
sentinel5p	s5p_aer_ai_granule_available	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
sentinel5p	s5p_ch4_granule_available	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
sentinel5p	s5p_co_granule_available	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
sentinel5p	s5p_hcho_granule_available	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
sentinel5p	s5p_no2_granule_available	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
sentinel5p	s5p_o3_granule_available	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
sentinel5p	s5p_so2_granule_available	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
tceq	co	06-14 00Z 0.1833, 06Z 0.1556, 12Z 0.2111, 18Z 0.1933, 06-15 00Z 0.2167, 06Z 0.1778, 12Z 0.2333, 18Z 0.2667, 06-16 00Z 0.2778, 06Z 0.2278, 12Z 0.2556, 18Z 0.2444, 06-17 00Z 0.2267
tceq	no2	06-14 00Z 1.929, 06Z 2.07, 12Z 2.26, 18Z 1.408, 06-15 00Z 2.617, 06Z 2.791, 12Z 5.438, 18Z 7.302, 06-16 00Z 10.67, 06Z 8.838, 12Z 5.771, 18Z 6.854, 06-17 00Z 7.274
tceq	so2	06-14 00Z -0.15, 06Z -0.2412, 12Z -0.2333, 18Z -0.1278, 06-15 00Z -0.1833, 06Z -0.2389, 12Z -0.2111, 18Z -0.07222, 06-16 00Z 0.01667, 06Z -0.09444, 12Z -0.08333, 18Z -0.07222, 06-17 00Z -0.006667

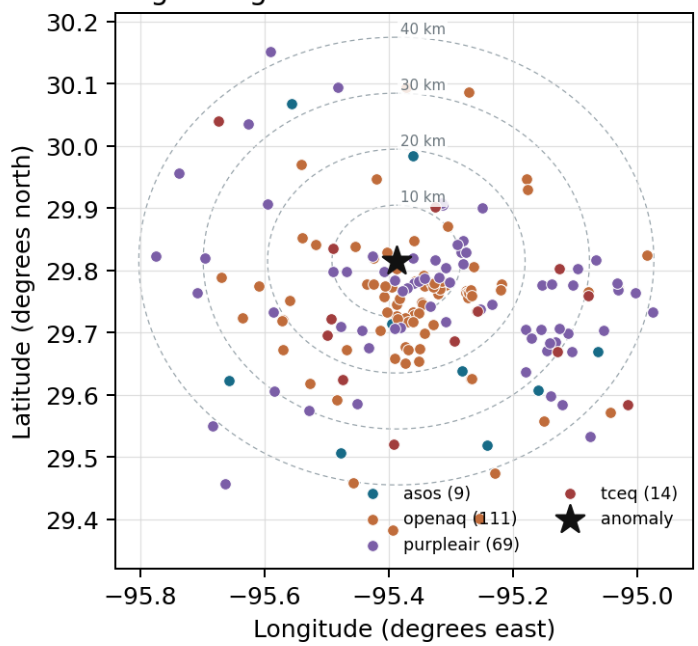
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

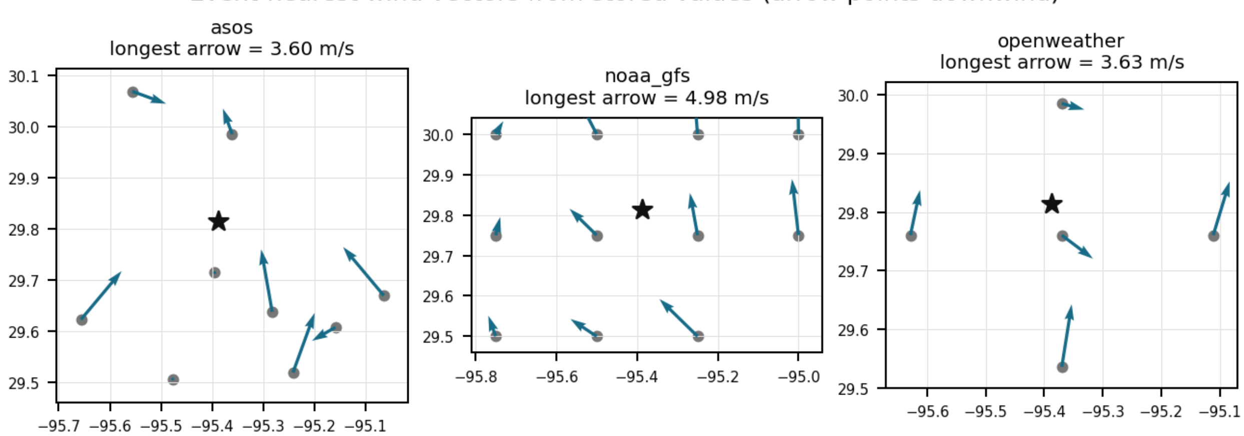
Source	Metric	Values
asos	humidity	MCJ (11.201 km) 91.43, IAH (19.067 km) 90.35, HOU (22.155 km) 86.74, EFD (31.938 km) 85.86, DWH (32.527 km) 88.46, SGR (33.627 km) 91.58, T41 (35.174 km) 86.25, AXH (35.363 km) 85.73, +1 more
asos	temperature	MCJ (11.201 km) 26.61, IAH (19.067 km) 26.72, HOU (22.155 km) 27.37, EFD (31.938 km) 27.64, DWH (32.527 km) 26.65, SGR (33.627 km) 26.74, T41 (35.174 km) 27.53, AXH (35.363 km) 26.78, +1 more
asos	wind_direction	MCJ (11.201 km) 127.9, IAH (19.067 km) 137.5, HOU (22.155 km) 141, EFD (31.938 km) 112.3, DWH (32.527 km) 114.4, SGR (33.627 km) 130.7, T41 (35.174 km) 152.6, AXH (35.363 km) 63.75, +1 more
asos	wind_speed	MCJ (11.201 km) 3.644, IAH (19.067 km) 2.229, HOU (22.155 km) 2.972, EFD (31.938 km) 3.416, DWH (32.527 km) 1.622, SGR (33.627 km) 2.687, T41 (35.174 km) 3.48, AXH (35.363 km) 0.9646, +1 more
noaa_gfs	gh_500	gfs:29.75,-95.50 (12.999 km) 5897, gfs:29.75,-95.25 (15.102 km) 5897, gfs:30.00,-95.50 (23.292 km) 5897, gfs:30.00,-95.25 (24.525 km) 5897, gfs:29.75,-95.75 (35.695 km) 5897, gfs:29.50,-95.50 (36.619 km) 5898, gfs:29.50,-95.25 (37.419 km) 5898, gfs:29.75,-95.00 (38.097 km) 5897, +3 more
noaa_gfs	pbl_height	gfs:29.75,-95.50 (12.999 km) 447.7, gfs:29.75,-95.25 (15.102 km) 434.3, gfs:30.00,-95.50 (23.292 km) 480.7, gfs:30.00,-95.25 (24.525 km) 436.5, gfs:29.75,-95.75 (35.695 km) 342, gfs:29.50,-95.50 (36.619 km) 351.4, gfs:29.50,-95.25 (37.419 km) 381.1, gfs:29.75,-95.00 (38.097 km) 414.5, +3 more
noaa_gfs	precipitable_water	gfs:29.75,-95.50 (12.999 km) 59.55, gfs:29.75,-95.25 (15.102 km) 59.92, gfs:30.00,-95.50 (23.292 km) 59.03, gfs:30.00,-95.25 (24.525 km) 59.42, gfs:29.75,-95.75 (35.695 km) 59.82, gfs:29.50,-95.50 (36.619 km) 60.06, gfs:29.50,-95.25 (37.419 km) 59.77, gfs:29.75,-95.00 (38.097 km) 59.79, +3 more
noaa_gfs	surface_pressure	gfs:29.75,-95.50 (12.999 km) 1009, gfs:29.75,-95.25 (15.102 km) 1011, gfs:30.00,-95.50 (23.292 km) 1008, gfs:30.00,-95.25 (24.525 km) 1009, gfs:29.75,-95.75 (35.695 km) 1008, gfs:29.50,-95.50 (36.619 km) 1010, gfs:29.50,-95.25 (37.419 km) 1011, gfs:29.75,-95.00 (38.097 km) 1012, +3 more
noaa_gfs	t_850	gfs:29.75,-95.50 (12.999 km) 18.44, gfs:29.75,-95.25 (15.102 km) 18.49, gfs:30.00,-95.50 (23.292 km) 18.41, gfs:30.00,-95.25 (24.525 km) 18.4, gfs:29.75,-95.75 (35.695 km) 18.46, gfs:29.50,-95.50 (36.619 km) 18.4, gfs:29.50,-95.25 (37.419 km) 18.51, gfs:29.75,-95.00 (38.097 km) 18.51, +3 more
noaa_gfs	u_10m	gfs:29.75,-95.50 (12.999 km) -0.9111, gfs:29.75,-95.25 (15.102 km) -0.5877, gfs:30.00,-95.50 (23.292 km) -0.6169, gfs:30.00,-95.25 (24.525 km) -0.5436, gfs:29.75,-95.75 (35.695 km) -0.7277, gfs:29.50,-95.50 (36.619 km) -1.159, gfs:29.50,-95.25 (37.419 km) -1.028, gfs:29.75,-95.00 (38.097 km) -0.4586, +3 more
noaa_gfs	v_10m	gfs:29.75,-95.50 (12.999 km) 2.136, gfs:29.75,-95.25 (15.102 km) 2.101, gfs:30.00,-95.50 (23.292 km) 2.29, gfs:30.00,-95.25 (24.525 km) 2.299, gfs:29.75,-95.75 (35.695 km) 2.189, gfs:29.50,-95.50 (36.619 km) 2.36, gfs:29.50,-95.25 (37.419 km) 2.57, gfs:29.75,-95.00 (38.097 km) 2.417, +3 more
openaq	ozone	1319333 (4.693 km) 0.01524, 319 (10.034 km) 0.01006, 2046 (10.097 km) 0.009759, 310 (11.311 km) 0.01414, 3378 (15.436 km) 0.009182, 325 (16.839 km) 0.0128, 2039 (16.93 km) 0.0114, 311 (17.034 km) 0.01433, +10 more
openaq	pm10	15852419 (8.427 km) 29.59, 13380082 (15.436 km) 31.89, 271 (29.738 km) 22.51
openaq	pm25	2071327 (0.0 km) 12.55, 10107877 (1.388 km) 10.77, 13787197 (2.124 km) 7.912, 8967015 (3.659 km) 5.672, 12296914 (4.519 km) 7.561, 13787194 (4.585 km) 7.724, 8967373 (4.84 km) 7.606, 8967193 (4.94 km) 8.222, +82 more
openweather	cloud_cover	grid:center (6.262 km) 84.73, grid:north (19.034 km) 83.81, grid:west (24.006 km) 87.2, grid:east (27.362 km) 83.39, grid:south (31.042 km) 83.17
openweather	humidity	grid:center (6.262 km) 90.97, grid:north (19.034 km) 90.46, grid:west (24.006 km) 90.5, grid:east (27.362 km) 89.83, grid:south (31.042 km) 87.49
openweather	precipitation	grid:center (6.262 km) 0.9068, grid:north (19.034 km) 1.074, grid:west (24.006 km) 1.127, grid:east (27.362 km) 1.35, grid:south (31.042 km) 1.029
openweather	pressure	grid:center (6.262 km) 1012, grid:north (19.034 km) 1012, grid:west (24.006 km) 1012, grid:east (27.362 km) 1012, grid:south (31.042 km) 1012
openweather	temperature	grid:center (6.262 km) 27.14, grid:north (19.034 km) 26.9, grid:west (24.006 km) 26.78, grid:east (27.362 km) 27.33, grid:south (31.042 km) 27.17
openweather	wind_direction	grid:center (6.262 km) 161.5, grid:north (19.034 km) 164, grid:west (24.006 km) 162.4, grid:east (27.362 km) 170.9, grid:south (31.042 km) 159.1
openweather	wind_speed	grid:center (6.262 km) 1.585, grid:north (19.034 km) 1.889, grid:west (24.006 km) 1.495, grid:east (27.362 km) 2.584, grid:south (31.042 km) 2.546

Source	Metric	Values
purpleair	pm25	246011 (2.65 km) 8.393, 223813 (2.654 km) 10.25, 302037 (3.453 km) 10.69, 155367 (3.78 km) 10.5, 127451 (4.849 km) 4.262, 186303 (5.082 km) 8.771, 27821 (5.158 km) 0.3219, 296101 (5.395 km) 10.75, +61 more
tceq	co	482011052 (0.02 km) 0.4, 482011035 (15.439 km) 0.1352, 482011039 (29.754 km) 0.1375
tceq	no2	482011052 (0.02 km) 14.98, 482010047 (10.026 km) 7.762, 482010024 (11.303 km) 1.856, 482011066 (14.47 km) 7.539, 482011035 (15.439 km) 8.625, 482010416 (16.848 km) 5.6, 482010055 (17.04 km) 5.301, 482010026 (25.33 km) 7.069, +5 more
tceq	so2	482011035 (15.439 km) -0.4178, 482010051 (22.781 km) 0.006849, 482011039 (29.754 km) 0.02083

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). The labeling guide that comes with this packet has the definitions and marking rules. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Claim 1 of 36

Relative humidity reached 100 percent around the time of the anomaly according to ASOS and OpenWeather observations, facilitating hygroscopic aerosol swelling or localized measurement distortion.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 2 of 36

A localized accumulation mechanism is supported because humidity near the anomaly reached 100 percent at 15:55 UTC, nearby wind speed dropped to 0.0 m/s at 15:55 UTC, and regional boundary-layer height was only about 187 to 238 m around 12 to 18 UTC, all of which favor trapping and amplification of near-surface particles in central Houston.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 3 of 36

Calm surface wind speeds recorded by ASOS stations and low planetary boundary layer height modeled by GFS produced stagnant atmospheric dispersion conditions that limited pollutant ventilation.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 4 of 36

The local Houston meteorology at the anomaly time favored aerosol stagnation and hygroscopic growth because humidity was near saturation and surface winds were very weak while boundary-layer depth was low.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 5 of 36

Ambient relative humidity reaching 100 percent in ASOS and OpenWeather data, combined with much lower PM2.5 concentrations at nearby PurpleAir sensors, supports hygroscopic particle growth or optical sensor humidity interference.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 6 of 36

The openaq central Houston PM2.5 anomaly is most likely a monitor-scale artifact or humidity-amplified local spike rather than a broad regional particulate event.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 7 of 36

The air quality anomaly is related to particulate matter (PM2.5) with a concentration of 5.6 ug/m3.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 8 of 36

Stagnant surface winds recorded by ASOS stations and a low planetary boundary layer height simulated by GFS impaired atmospheric dispersion, promoting the localized accumulation of PM2.5 observed by OpenAQ.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 9 of 36

The anomaly occurred on June 15th, 2026, at approximately 16:00:00 UTC.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 10 of 36

A monitor-specific artifact or humidity-driven overresponse remains credible because the anomalous openaq PM2.5 value was 52.0 ug/m3 at the anomaly location while a nearby purpleair PM2.5 sensor measured only 5.6 ug/m3 at the same time and nearby openaq PM10 was only 31.0 ug/m3, which is weak corroboration for a severe area-wide fine-particle event.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 11 of 36

The location of the anomaly is in the vicinity of grid:center (6.262 km) with a distance of approximately 2.65 km from the nearest sensor.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 12 of 36

Meteorological stagnation and moisture trapping are a plausible cause of elevated local PM2.5 because winds near the anomaly were very weak to calm, relative humidity was about 95 to 100 percent, cloud cover was near 100 percent, precipitable water was high, and modeled boundary-layer height was low, all of which favor poor dispersion and aerosol accumulation in Houston.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 13 of 36

The available co-pollutant pattern does not support a strong combustion or large industrial plume at the anomaly time because nearby ozone was low, PM10 was only moderate, and co-located CO and NO2 were not sharply elevated.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 14 of 36

The anomaly is caused by a nearby industrial facility releasing excessive amounts of particulate matter into the atmosphere.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 15 of 36

The openaq PM2.5 measurement at 29.815, -95.388 in Houston was 52.0 ug/m3 at 2026-06-15T16:00:00+00:00, compared with an expected value of 10.268711656441717 ug/m3, giving a z-score of 9.608890843426831 and severe anomaly classification.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 16 of 36

Hygroscopic particle growth or PM2.5 measurement bias under saturated conditions is a plausible contributor because the anomaly occurred during extremely high humidity and light precipitation conditions, while regional PM2.5 and PM10 did not show a comparably strong increase and colocated combustion tracers were not sharply elevated.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 17 of 36

The nearest contemporaneous gas measurements were openaq ozone at 0.011 ppm 10.034 km from the anomaly location, tceq CO at 0.4 ppm 0.02 km from the anomaly location, tceq NO2 at 7.2 ppb 10.026 km from the anomaly location, and tceq SO2 at -0.5 ppb 15.439 km from the anomaly location, all at 2026-06-15T16:00:00+00:00.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 18 of 36

A TCEQ monitor situated 0.02 km from the anomaly location recorded a co-located carbon monoxide concentration of 0.4 ppm at 2026-06-15T16:00:00+00:00.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 19 of 36

The air quality anomaly in Houston, Texas on June 15th, 2026 was also characterized by elevated levels of NO2.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 20 of 36

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5 particles.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 21 of 36

The anomaly is a result of a combination of factors including high temperatures, low humidity, and stagnant atmospheric conditions, leading to poor air mixing and increased pollutant concentrations.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 22 of 36

A localized primary emission source near the sensor elevated PM2.5 concentrations measured by OpenAQ alongside concurrently elevated carbon monoxide concentrations recorded by TCEQ.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 23 of 36

The nearest openaq PM10 measurement was 31.0 ug/m3 at 2026-06-15T16:00:00+00:00 and 8.427 km from the anomaly location.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 24 of 36

A localized near-source particulate plume is a plausible cause of the anomalous PM2.5 measurement because the anomalous openaq sensor reported 52.0 ug/m3 while nearby purpleair PM2.5 and nearby openaq PM10 remained much lower at the same time, which is consistent with a very spatially confined source affecting one monitor more than the surrounding network.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 25 of 36

A severe thunderstorm or tornado event occurred in the area, stirring up pollutants and reducing air quality.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 26 of 36

At the time of the peak PM2.5 observation on June 15, 2026, at 16:00 UTC, the nearest PurpleAir sensor situated 2.65 km away recorded a PM2.5 concentration of 5.6 ug/m3.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 27 of 36

The air quality anomaly in Houston, Texas on June 15th, 2026 was characterized by elevated levels of PM2.5.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 28 of 36

Near-saturated relative humidity and precipitation recorded by OpenWeather and ASOS promoted hygroscopic growth of fine particles or created optical scattering artifacts on the OpenAQ PM2.5 sensor.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 29 of 36

Elevated carbon monoxide and nitrogen dioxide concentrations recorded at the nearest TCEQ monitor support localized primary combustion or industrial emissions contributing to the PM2.5 elevation.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 30 of 36

An OpenAQ sensor in Houston recorded a PM2.5 concentration spike of 52.0 ug/m3 at 16:00 UTC on June 15, 2026, whereas nearby OpenAQ and PurpleAir sensors indicated low background PM2.5 levels.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 31 of 36

The air quality anomaly in Houston, Texas on June 15th, 2026 was likely caused by a combination of factors including weather patterns and local emissions.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 32 of 36

Meteorological observations from ASOS and GFS atmospheric model outputs indicated surface stagnation near the anomaly site, featuring calm surface winds and reduced boundary layer heights.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 33 of 36

The air quality anomaly in Houston, Texas is caused by high levels of NO2 particles.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 34 of 36

A strong regional combustion or petrochemical plume is not well supported because ozone near the anomaly was only 0.011 ppm, nearby TCEQ CO was 0.4 ppm, nearby TCEQ NO2 was 7.2 ppb, nearby TCEQ SO2 was negative to near zero, and the available Sentinel-5P trace-gas columns in the context do not show an obvious broad elevated signal near the anomaly time.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 35 of 36

The air quality anomaly in Houston, Texas is caused by high levels of CO particles.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 36 of 36

An OpenAQ monitoring sensor located at (29.815, -95.388) in Houston recorded a PM2.5 concentration of 52.0 ug/m3 at 2026-06-15T16:00:00+00:00, exceeding the expected baseline of 10.27 ug/m3 with a z-score of 9.61.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.